Year 11 Higher

REVISION PACK

Graphs 5

Name: _____

Class: _____

Graphs 5 · Simultaneous equations (Lesson 3.1)

Worked example: adding to remove a letter

Solve the simultaneous equations

$2x + 3y = 12$

$4x - 3y = 6$

Step 1	$+3y$ and $-3y$: add	$6x = 18$
Step 2	Divide by 6	$x = 3$
Step 3	Put x in	$2(3) + 3y = 12$
Step 4	Subtract 6, divide by 3	$y = 2$
Step 5	Answer	$x = 3,\ y = 2$

Practice: adding to remove a letter

1.1 Solve the simultaneous equations

$4x + 3y = 18$

$2x - 3y = 0$

1.2 Solve the simultaneous equations

$5x + 2y = 18$

$3x - 2y = -2$

1.3 Solve the simultaneous equations

$2x + 5y = 13$

$4x - 5y = 11$

1.4 Solve the simultaneous equations

$3x + 2y = 18$

$x - 2y = -2$

1.5 Solve the simultaneous equations

$6x + y = 11$

$2x - y = -3$

1.6 Solve the simultaneous equations

$5x + 4y = 6$

$3x - 4y = 10$

Graphs 5 · Simultaneous equations (Lesson 3.1)

Worked example: multiplying first

Solve the simultaneous equations

$3x + 2y = 12$

$2x - 3y = -5$

Step 1	First × 3: same y number	$9x + 6y = 36$
Step 2	Second × 2: same y number	$4x - 6y = -10$
Step 3	Add to remove y	$13x = 26$
Step 4	Divide by 13	$x = 2$
Step 5	Put x in	$3(2) + 2y = 12$
Step 6	Subtract 6, divide by 2	$y = 3$

Practice: multiplying first

1.7 Solve the simultaneous equations

$5x + 3y = 12$

$2x - 4y = 10$

1.8 Solve the simultaneous equations

$3x + 4y = 10$

$5x - 3y = 7$

1.9 Solve the simultaneous equations

$2x + 5y = 13$

$3x + 2y = 3$

1.10 Solve the simultaneous equations

$4x + 5y = 13$

$3x - 2y = 4$

1.11 Solve the simultaneous equations

$6x + 5y = 16$

$4x - 3y = -2$

1.12 Solve the simultaneous equations

$3x + 2y = 11$

$4x - 5y = 7$

Graphs 5 · Simultaneous equations (Lesson 3.1)

Worked example: problems with two unknowns

4 pens and 1 ruler cost 2.100 KD. 2 pens and 3 rulers cost 1.800 KD. Work out the cost of one pen and the cost of one ruler.

Step 1	Equation 1	$4p + r = 2.1$
Step 2	Second × 2: same p number	$4p + 6r = 3.6$
Step 3	Subtract to remove p	$5r = 1.5$
Step 4	Divide by 5	$r = 0.3$
Step 5	Put r in	$4p + 0.3 = 2.1$
Step 6	Subtract 0.3, divide by 4	$p = 0.45$
Step 7	Answer	$\text{pen } 0.450 \text{ KD},\ \text{ruler } 0.300 \text{ KD}$

Practice: problems with two unknowns

1.13 2 notebooks and 3 pencils cost 2.250 KD. 1 notebook and 4 pencils cost 1.750 KD. Work out the cost of one notebook and the cost of one pencil.

1.14 5 apples and 2 mangos cost 3.300 KD. 3 apples and 4 mangos cost 3.800 KD. Work out the cost of one apple and the cost of one mango.

1.15 3 coffees and 1 cake cost 4.100 KD. 2 coffees and 5 cakes cost 4.900 KD. Work out the cost of one coffee and the cost of one cake.

1.16 2 sandwiches and 1 juice cost 3.500 KD. 1 sandwich and 3 juices cost 3.000 KD. Work out the cost of one sandwich and the cost of one juice.

1.17 4 pencils and 3 erasers cost 2.800 KD. 2 pencils and 5 erasers cost 3.500 KD. Work out the cost of one pencil and the cost of one eraser.

1.18 3 books and 2 magazines cost 10.000 KD. 1 book and 2 magazines cost 5.000 KD. Work out the cost of one book and the cost of one magazine.

Graphs 5 · Simultaneous equations (Lesson 3.1)

Worked example: adding the equations

Solve the simultaneous equations

$4x + y = 11$

$2x - y = 1$

Step 1	$+y$ and $-y$: add	$6x = 12$
Step 2	Divide by 6	$x = 2$
Step 3	Put x in	$4(2) + y = 11$
Step 4	Subtract 8	$y = 3$

Practice: adding the equations

1.19 Solve the simultaneous equations

$3x + y = 7$

$2x - y = 3$

1.20 Solve the simultaneous equations

$4x + y = 11$

$x - y = -1$

1.21 Solve the simultaneous equations

$2x + y = 8$

$5x - y = 13$

1.22 Solve the simultaneous equations

$x + y = 5$

$3x - y = 11$

1.23 Solve the simultaneous equations

$5x + y = 9$

$2x - y = -2$

1.24 Solve the simultaneous equations

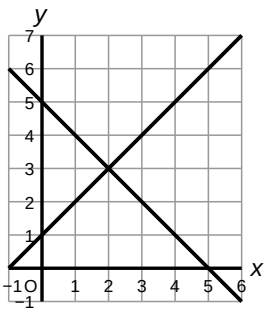
$3x + y = 11$

$4x - y = 3$

Graphs 5 · Simultaneous equations (Lesson 3.1)

Worked example: solving using graphs

Draw the graphs of $y = x + 1$ and $x + y = 5$. Use your graphs to solve the simultaneous equations.



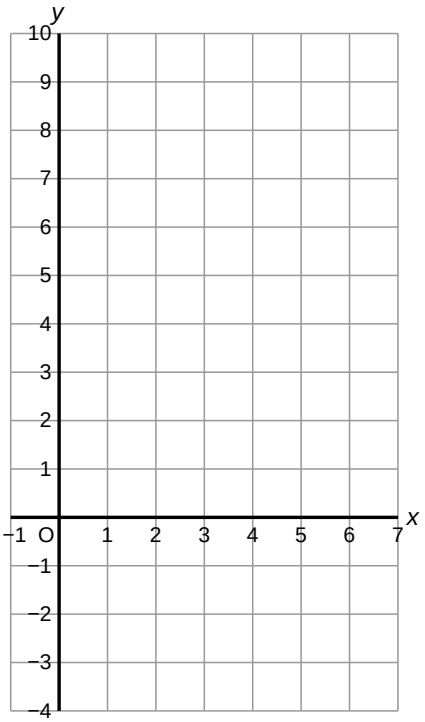
Step 1	Draw $y = x + 1$	plot $(0, 1)$ and $(4, 5)$ and join
Step 2	Draw $x + y = 5$	plot $(0, 5)$ and $(5, 0)$ and join
Step 3	Read the crossing point	$(2, 3)$
Step 4	Answer	$x = 2$, $y = 3$

Practice: solving using graphs

2.1 (a) On the grid, draw the graph of $y = 2x - 3$ for values of x from -1 to 5

(b) On the same grid, draw the graph of $x + y = 6$

(c) Use your graphs to solve the simultaneous equations $y = 2x - 3$ and $x + y = 6$

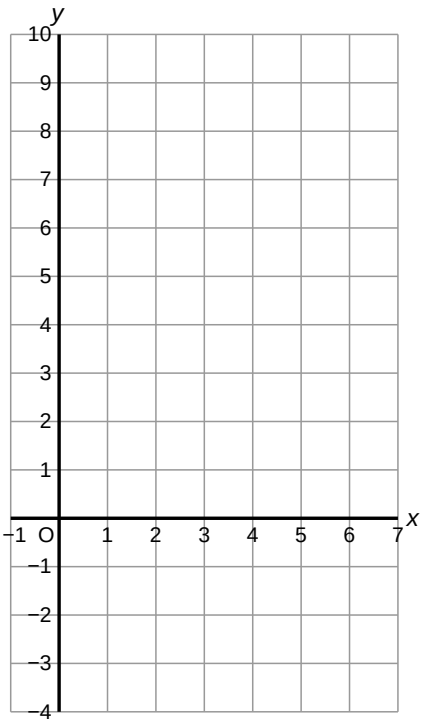


Practice: solving using graphs (continued)

2.2 (a) On the grid, draw the graph of $y = x + 2$ for values of x from -1 to 6

(b) On the same grid, draw the graph of $y = 8 - x$

(c) Use your graphs to solve the simultaneous equations $y = x + 2$ and $y = 8 - x$

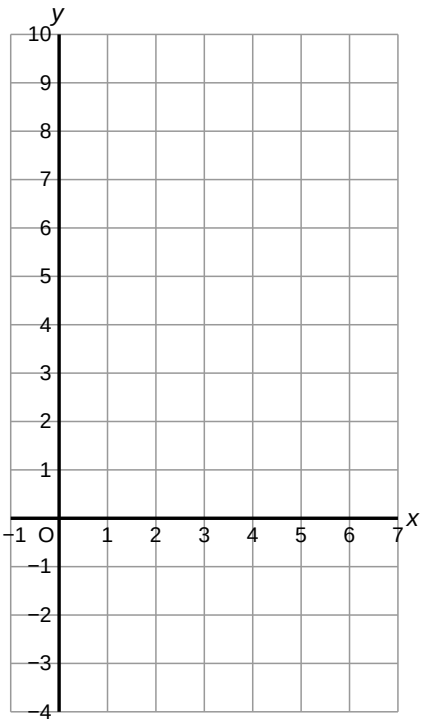


Practice: solving using graphs (continued)

2.3 (a) On the grid, draw the graph of $y = 3x - 2$ for values of x from 0 to 4

(b) On the same grid, draw the graph of $x + y = 6$

(c) Use your graphs to solve the simultaneous equations $y = 3x - 2$ and $x + y = 6$



Practice: solving using graphs (continued)

2.4 (a) On the grid, draw the graph of $y = 2x + 1$ for values of x from -1 to 4

(b) On the same grid, draw the graph of $y = 7 - x$

(c) Use your graphs to solve the simultaneous equations $y = 2x + 1$ and $y = 7 - x$

